

Data Analysis with R and Python

Semestral Examination

22 MAY 2026, 4:00PM – 5:30PM

Bachelor of Statistical Data
Science

TOTAL POINTS: 25

PART B INSTRUCTIONS

There are a total of 29 points on this paper. The maximum you can score is 25.

Write your answers on the question paper. Write only the answers to the questions asked. Do not write any R code or any other explanation unless instructed.

This is a practical examination; you may use laptops or desktops with R and necessary packages pre-installed, as per instructions given earlier. However, you are not allowed to access the internet during the examination.

1. Indicate whether the following statements are TRUE or FALSE. [5]

- (a) In a standard boxplot, the central line represents the sample mean.
TRUE ☐ FALSE ☐
- (b) A scatter plot is generally the most appropriate visualization for showing the relationship between two categorical variables.
TRUE ☐ FALSE ☐
- (c) In R, a list is a generic vector that can contain objects of different types, while an atomic vector is restricted to elements of a single type.
TRUE ☐ FALSE ☐
- (d) In a distribution that is significantly skewed to the right (positively skewed), the mean is typically smaller than the median.
TRUE ☐ FALSE ☐
- (e) A histogram can be viewed as an estimator of the probability density function of the underlying distribution.
TRUE ☐ FALSE ☐

For the remaining questions, consider the UN dataset in the `carData` package in R. Take the subset of rows excluding those which have missing values for either `pctUrban` or `infantMortality`. The resulting dataset should have 193 rows.

We are interested in modeling the conditional distribution of y given x , where y represents `infantMortality` and x represents `pctUrban`. We consider the following linear regression models as potential choices:

- Model 1: $y = a_1 + b_1x + \varepsilon$
- Model 2: $\log y = a_2 + b_2x + \varepsilon$
- Model 3: $\log y = a_3 + b_3\sqrt{x} + \varepsilon$

Report all numeric answers rounded to 3 places after the decimal.

2. Compute and report the sample correlation coefficients for the three models. [3]

- Model 1: $\text{Cor}(x, y) = \underline{\hspace{2cm}}$
- Model 2: $\text{Cor}(x, \log y) = \underline{\hspace{2cm}}$
- Model 3: $\text{Cor}(\sqrt{x}, \log y) = \underline{\hspace{2cm}}$

3. Suppose we estimate each model using the least squares method. The resulting estimates are: [3]

- Model 1: $\hat{a}_1 = \underline{\hspace{2cm}}$ $\hat{b}_1 = \underline{\hspace{2cm}}$
- Model 2: $\hat{a}_2 = \underline{\hspace{2cm}}$ $\hat{b}_2 = \underline{\hspace{2cm}}$
- Model 3: $\hat{a}_3 = \underline{\hspace{2cm}}$ $\hat{b}_3 = \underline{\hspace{2cm}}$

4. Compute and report the corresponding minimum least squares value (also known as the “residual sum of squares” or RSS) for the three models. [3]

- Model 1: $\text{RSS} = \underline{\hspace{2cm}}$
- Model 2: $\text{RSS} = \underline{\hspace{2cm}}$
- Model 3: $\text{RSS} = \underline{\hspace{2cm}}$

5. Assuming that the errors ε are independent $N(0, \sigma^2)$ random variables for some unknown $\sigma^2 > 0$, compute the likelihood function for each model evaluated at the corresponding least squares estimates $\hat{a}_i, \hat{b}_i$ given above. Remember that the likelihood function is based on the density of y , even when the model is specified in terms of $\log y$. For the unknown σ^2 , use the sample variance of the corresponding residuals. [6]

- Model 1: likelihood evaluated at $\hat{a}_1, \hat{b}_1 =$ _____
- Model 2: likelihood evaluated at $\hat{a}_2, \hat{b}_2 =$ _____
- Model 3: likelihood evaluated at $\hat{a}_3, \hat{b}_3 =$ _____

6. Is it a reasonable strategy to choose a “best” model (out of these three) on the basis of correlation? Briefly justify your answer. [3]

7. Is it a reasonable strategy to choose a “best” model (out of these three) on the basis of RSS? Briefly justify your answer. [3]

8. Is it a reasonable strategy to choose a “best” model (out of these three) on the basis of likelihood? Briefly justify your answer. [3]